WATERTANK: SMART WATER TANK CONTROLLER AI AGENT

ASSIGNMENT: AI AGENT FORMULATION AND ANALYSIS

1. Agent Overview and Problem Identification

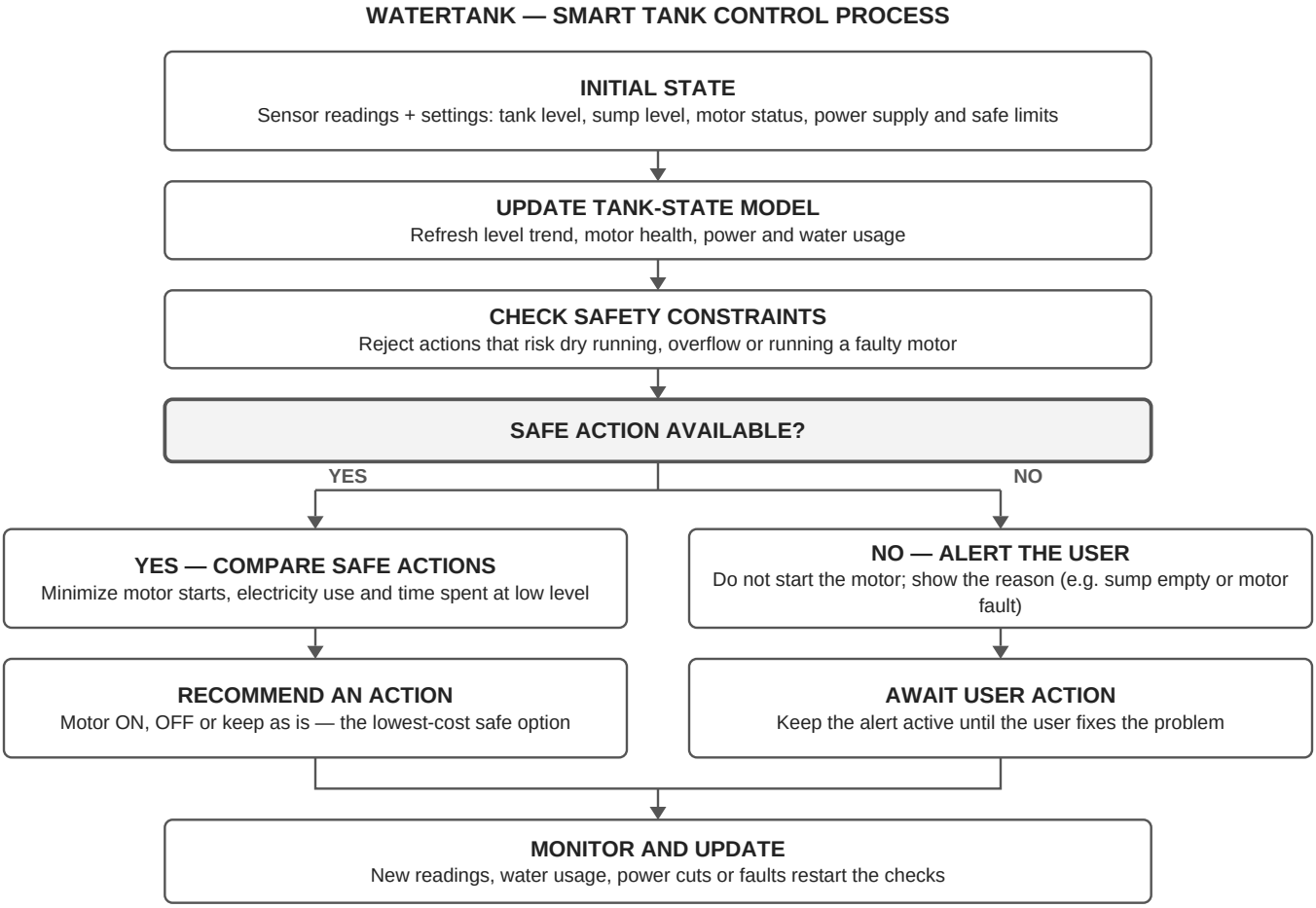
WaterTank is a decision-support AI agent that decides when a household water tank motor should be turned ON or OFF. It considers the tank water level, the source (sump) level, motor status and health, power availability, and usage patterns. Its goal is to keep the tank within a safe level range, avoid dry running and overflow, avoid unnecessary motor starts, and alert the user when the situation is unsafe.

2. Agent Type

WaterTank is a model-based, goal-based agent with a simple utility preference. It maintains an updated representation of the tank, motor and power conditions, rejects actions that violate safety constraints, and compares the remaining safe options. Among safe options, it prefers fewer motor starts, lower electricity use, and less time spent at a low water level.

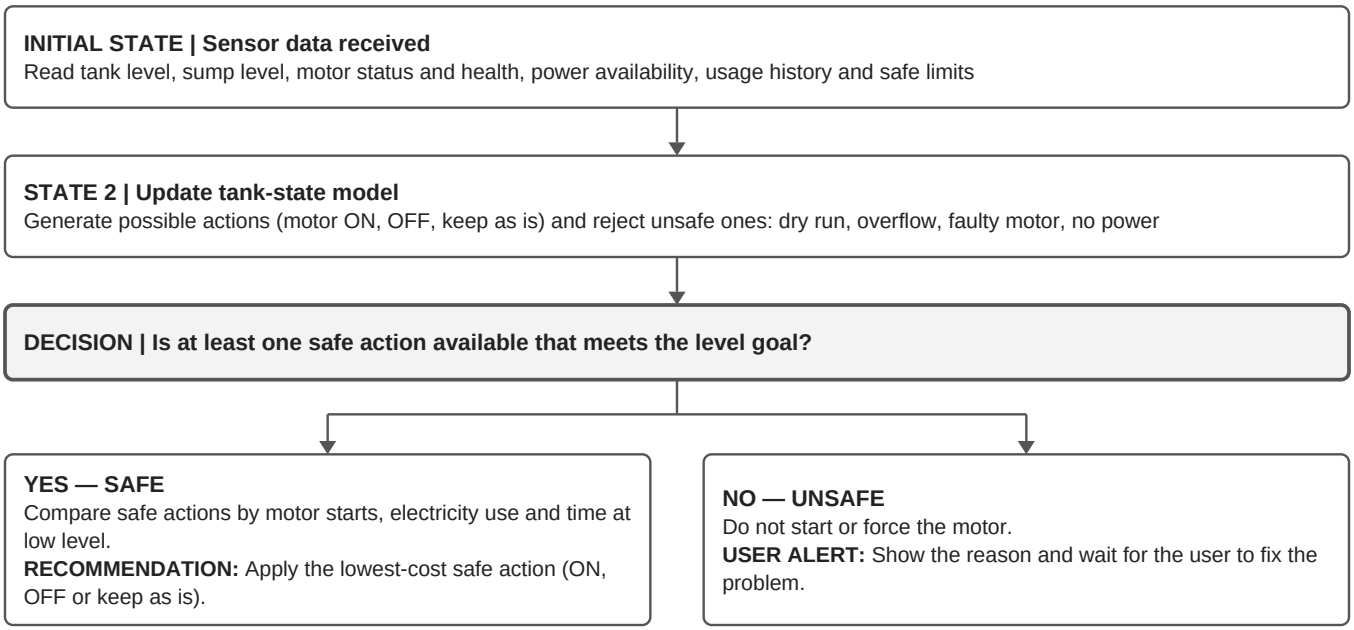
3. Graphical Representation

The top-down flowchart shows how WaterTank checks safety constraints, handles safe and unsafe cases, and revisits decisions when tank conditions change.



4. Problem Formulation

The agent checks hard safety constraints before comparing the cost of valid actions. If no safe action can meet the water-level goal, it does not force the motor to run.



FEEDBACK LOOP: New water usage, power cuts, motor faults or level changes update the model and restart the checks.

Operational boundary: WaterTank only switches or recommends the motor ON/OFF within safe limits. It does not repair equipment, fix leaks, or override the user's manual control.

5. PEAS Framework

PEAS component	WaterTank description
Performance measure	Keep the tank level in the safe range; avoid dry running and overflow; reduce unnecessary motor starts and electricity use; alert the user in unsafe cases.
Environment	Water tank, underground sump or source, motor/pump, power supply, pipes, and household water users.
Actuators	Motor ON/OFF command (or recommendation), buzzer or app alerts, level display, and usage reports.
Sensors	Tank level sensor, sump level sensor, motor status/temperature/current sensor, power sensor, and clock.

6. Environmental Properties

Property	Classification	Reason
Observability	Partially observable	Sensors can give wrong readings, and leaks or blockages may not be visible.
Determinism	Stochastic	Water usage, power cuts, and motor faults are unpredictable.
Episodes	Sequential	Running the motor now changes the water level available for later decisions.
Change	Dynamic	Tank level, sump level, and power change even while the agent is deciding.
Values	Hybrid	Motor ON/OFF is discrete; water level, time, and usage are numerical.
Agents	Single-agent	WaterTank is the only decision-maker; household usage and power cuts are treated as environment events.

7. Example Scenario

The tank level is 15% and the minimum safe level is 25%. The sump has 60% water, the motor is healthy, and power is available. WaterTank rejects "keep motor OFF" because the tank would soon run empty, checks that dry running, overflow and motor faults are not a risk, and recommends turning the motor ON until the level reaches 90%. If the sump were empty or the motor were faulty, no safe action would exist, so it would alert the user instead of starting the motor.

Conclusion

WaterTank is a model-based, goal-based decision-support agent for sequential water tank control. It prioritizes safe actions, accounts for changing and uncertain conditions, and leaves unresolved problems to the user.